Cell Biology: The Cell Cycle and Mitosis (Page 1)

1. Introduction to the Cell Cycle
The cell cycle is an ordered series of events involving cell growth and cell division that produces two new daught

2. Interphase
Interphase is the phase of the cell cycle in which a typical cell spends most of its life. During this phase, the cell
- G1 Phase (First Gap): The cell grows larger physically, copies organelles, and makes the molecular building bl
- S Phase (Synthesis): The cell synthesizes a complete copy of the DNA in its nucleus. It also duplicates a micro
- G2 Phase (Second Gap): The cell grows more, makes proteins and organelles, and begins to reorganize its co

Cell Biology: The Cell Cycle and Mitosis (Page 2)

3. Phases of Mitosis
Mitosis consists of four primary stages:
- Prophase: Chromatin condenses into visible chromosomes. The nucleolus disappears and the mitotic spindle b
- Metaphase: Chromosomes align along the metaphase plate (the equatorial plane of the cell). Each sister chrom
- Anaphase: Sister chromatids are pulled apart at the centromere by shortening spindle microtubules and move t
- Telophase: A nuclear membrane reforms around each set of separated chromosomes. Chromosomes begin to

4. Cytokinesis
Cytokinesis is the physical division of the cytoplasm into two daughter cells. In animal cells, cytokinesis occurs th